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# UNIT-3 : PROCESSING OF SMART MATERI

IMPORTANT: Syllabus: Semiconductors and their processing, Metals and metallization techniques, Ceramics and their processing, Polymers and their synthesis, UV radiation curing of polymers, fluids.

## 3.1 Semiconductors and their Processing

Definition: Semiconductors are materials whose electrical conductivity ($10^{-8}$ to $10^{3}$ S/cm) lies between conductors and insulators and can be controlled by doping, temperature, light or field. Silicon is the most important semiconductor for sensors, actuators and MEMS.

### Types

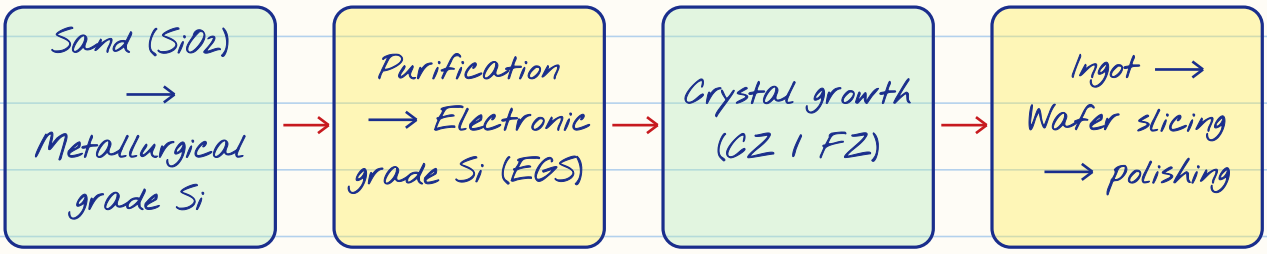
- Elemental : Silicon (Si), Germanium (Ge). Compound : GaAs, GaN, InP, SiC, ZnO, CdS.
- Intrinsic (pure) and extrinsic (doped): n-type (donor P, As, Sb - extra electrons) and p-type (acceptor B, Ga - holes).
- Band gap: Si = 1.12 eV, Ge = 0.67 eV, GaAs = 1.43 eV.

### Why silicon for smart systems?

- Abundant (sand), excellent mechanical properties (strong, elastic, no hysteresis), piezoresistive, stable oxide ($SiO_2$), well-developed IC technology, can integrate sensor + electronics on one chip.

### Steps in silicon wafer processing

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### (a) Raw material to electronic-grade silicon

- Sand reduced with carbon in an arc furnace at ~1800 C : $SiO_2 + 2C \rightarrow Si + 2CO$ (98% pure, MGS).
- MGS is converted into trichlorosilane ($SiHCl_3$) : $Si + 3HCl \rightarrow SiHCl_3 + H_2$ ; distilled to remove impurities.
- Siemens process : $SiHCl_3 + H_2 \rightarrow Si$ (polysilicon rods) + 3HCl at ~1100 C → 99.9999999% (9N) pure EGS.

### (b) Single-crystal growth

- Czochralski (CZ) method : polysilicon melted in a quartz crucible (1420 C); a seed crystal is dipped in the melt, slowly pulled upward (1-2 mm/min) while rotating → single-crystal ingot of 200-300 mm diameter. Dopants added in the melt.
- Float-zone (FZ) method : a molten zone is passed along a polysilicon rod using an RF coil; no crucible → higher purity , used for power devices and detectors; expensive, smaller diameter.

### (c) Wafer preparation

- Ingot trimmed, orientation flat ground, sliced by diamond saw (0.5-0.7 mm), lapped , etched (KOH / HF-$HNO_3$), and chemical-mechanical polished (CMP) to mirror finish.

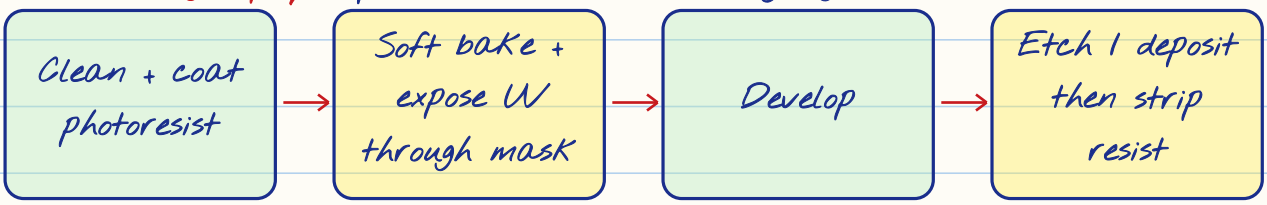
### (d) Microfabrication steps (IC and MEMS)

- Oxidation : thermal growth of $SiO_2$ layer at 900-1200 C in $O_2$ (dry) or steam (wet) - used as insulator, mask and passivation.
  - Dry: $Si + O_2 \rightarrow SiO_2$ (thin, high quality); Wet: $Si + 2H_2O \rightarrow$

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$SiO_2 + 2H_2$ (thick, fast).

- Photolithography : pattern transfer using light.




  - Positive resist : exposed part dissolves; negative resist : exposed part hardens (e.g. SU-8).
- Doping : by thermal diffusion (dopant at 900–1100 C, profile by Fick's law) or ion implantation (accelerated ions 10–500 KeV, precise dose, followed by annealing).
- Thin film deposition : CVD (LPCVD, PECVD), PVD (sputtering, evaporation), epitaxy.
- Etching : removal of material.
  - Wet etching : KOH, TMAH (anisotropic - along crystal planes, forms V-grooves, diaphragms), HF (isotropic).
  - Dry etching : plasma / RIE, DRIE (Bosch process) - deep, vertical trenches, high aspect ratio (MEMS).
- Bulk micromachining (etching the substrate) and surface micromachining (deposit and etch sacrificial layers) - used to make cantilevers, membranes and comb drives.
- Packaging : dicing, die bonding, wire bonding, encapsulation, testing.

Note: Wafer bonding (anodic, fusion) and LIGA (X-ray lithography + electroplating + moulding) are special MEMS techniques for high-aspect-ratio structures.

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## 3.2 Metals and Metallization Techniques

**Definition:** Metallization is the process of depositing thin metal films on a semiconductor / ceramic / polymer substrate to make electrical interconnects, contacts, electrodes, bond pads and mirrors.

### Metals used

| Metal | Use | Remark |
|---|---|---|
| Aluminium (Al) | Interconnect, bond pads | Low cost, good adhesion to $SiO_2$, electromigration |
| Copper (Cu) | Advanced interconnect | Low resistivity, needs barrier (TaN), damascene process |
| Gold (Au) | Contacts, wire bonds, electrodes | Inert, corrosion free, costly |
| Platinum (Pt) | Electrodes (PZT films), heaters | High temperature stable |
| Titanium / Chromium | Adhesion layers | Thin (10–50 nm) glue layer |
| Tungsten (W) | Vias, plugs | Refractory, CVD |
| Silicides ($TiSi_2$, $PtSi$) | Ohmic contacts, gates | Low resistance contact to Si |

### Metallization techniques

#### 1. Physical Vapour Deposition (PVD)

- **Thermal / e-beam evaporation** : metal heated in vacuum ($10^{-6}$ torr) until it evaporates and condenses on the cooled substrate; e-beam heats high-melting metals. Line-of-sight, poor step coverage.
- **Sputtering** : $Ar^+$ ions from plasma bombard a metal target ,

ejecting atoms that deposit on the wafer. DC (metals), RF (insulators), magnetron sputtering (high rate). Good adhesion, good step coverage, alloys possible.

### 2. Chemical Vapour Deposition (CVD)

- Gaseous precursors react on a heated substrate to form a film (e.g. $WF_6 + 3H_2 \rightarrow W + 6HF$). Excellent conformal coverage; used for W, polysilicon, $SiO_2$, $Si_3N_4$.

### 3. Electroplating / Electroless plating

- Electroplating : substrate acts as cathode in a metal-salt bath; current deposits metal (Cu, Ni, Au). Used to grow thick metal structures ( LIGA , Cu damascene).
- Electroless plating : autocatalytic chemical reduction, no external current (Ni-P, Cu) - uniform on insulators.

### 4. Others

- Screen printing / thick-film (Ag, Ag-Pd pastes on ceramics - electrodes for PZT), spray / dip coating , cladding , hot dipping (galvanising) , ion plating .

### Patterning of metal films

- Subtractive (etch-back) : deposit blanket metal → photolithography → wet / dry etch metal.
- Lift-off : pattern resist first, deposit metal, dissolve resist - metal on resist lifts off (used for Au, Pt hard to etch).
- Damascene (Cu) : etch trenches in dielectric, deposit barrier + Cu seed, electroplate Cu, CMP excess.

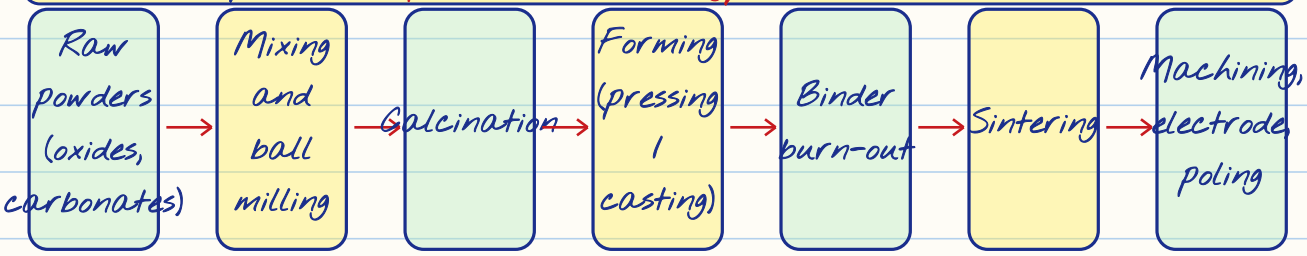
### Important issues

- Adhesion (Ti/Cr layer), contact resistance , electromigration (Al wires), step coverage , stress in film, diffusion barrier (TiN, TaN), annealing / sintering of contacts at 400-450 C.

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## 3.3 Ceramics and their Processing

Definition: Processing of ceramics: powder preparation → mixing / milling → forming → drying → sintering → finishing (electroding, poling). Ceramics cannot be melted and cast easily, so the powder metallurgy route is used.




### Step by step (for PZT / piezo ceramic)

- 1. Powder preparation : solid-state (mixed-oxide) route - weigh PbO, $ZrO_2$, $TiO_2$ in correct ratio; ball mill (wet in alcohol / water) to mix and reduce particle size.
  - Wet chemical routes : sol-gel , co-precipitation, hydrothermal, combustion - give fine, homogeneous, high-purity powders at lower temperature.
- 2. Calcination : heating the powder (800-900 C, 2-4 h) so that the oxides react to form the perovskite phase; powder is again milled (sub-micron).
- 3. Binder addition & granulation : PVA binder (2-5%) is added to give green strength.
- 4. Forming (shaping) of green body :
  - Dry (uniaxial) pressing - die pressing, 50-150 MPa, simple shapes (discs, plates).
  - Cold isostatic pressing (CIP) - uniform pressure by fluid, uniform density.
  - Tape casting (doctor blade) - thin sheets (20-200 micron) for

multilayer actuators and capacitors.

- Extrusion, slip casting (slurry poured into plaster mould), injection moulding (complex parts).
- Screen printing / spin coating - thick / thin films.

- 5. Binder burn-out : slow heating to 400-600 C to remove organics.
- 6. Sintering : heating at 1100-1300 C (below melting) for 1-4 hours: particles bond by diffusion, pores shrink, grains grow, density approaches 95-99% (shrinkage 15-20%).
  - PbO is volatile → sinter in a PbO-rich atmosphere (sealed crucible with $PbZrO_3$ powder).
  - Types: solid-state sintering, liquid-phase sintering, hot pressing, spark plasma sintering (SPS) - faster, finer grains.
- 7. Finishing : grinding, lapping, cutting to size.
- 8. Electroding : silver paste fired at ~ 600-800 C or sputtered Au / Pt / Ni.
- 9. Poling : apply DC field (2-4 KV/mm) in silicone oil bath at 100-150 C for 5-30 min.

IMPORTANT: Multilayer actuators (MLA) are made by tape casting + screen printing electrodes + stacking + co-firing (Ag-Pd internal electrodes).

Properties controlled by processing

- Density, grain size, porosity, purity, phase composition ( morphotropic phase boundary, Zr/Ti ~ 52/48 ) decide d33, K, and dielectric constant.

Thin film ceramic (for MEMS)

- Sol-gel spin coating, RF sputtering, pulsed laser deposition (PLD), MOCVD; annealing 600-700 C to crystallise PZT thin film (0.5-3 micron).

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### Defects in ceramics

- Cracks, warping, porosity, abnormal grain growth, lead loss, low density, inhomogeneous shrinkage.

## 3.4 Polymers and their Synthesis

Definition: Polymerisation is the chemical process by which small molecules (monomers) are joined to form long chain macromolecules (polymers). The two main mechanisms are addition (chain-growth) and condensation (step-growth) polymerisation.

### A) Addition (chain-growth) polymerisation

- Monomers with a double bond (ethylene, styrene, vinyl chloride, acrylates) add to a growing chain, no by-product.
- Three steps :
  - Initiation : initiator (peroxide, AIBN) $\longrightarrow$ free radical R*; $R^* + M \longrightarrow R\text{-}M^*$
  - Propagation : $R\text{-}M^* + M \longrightarrow R\text{-}M\text{-}M^*$ ... chain grows rapidly
  - Termination : by combination or disproportionation of two radicals.
- Types of chain growth: free radical, cationic, anionic, coordination (Ziegler-Natta).
- Eg: polyethylene, PVC, polystyrene, PMMA, PVDF (from $VF_2$: $CH_2{=}CF_2$).

### B) Condensation (step-growth) polymerisation

- Monomers with two or more functional groups (-OH, -COOH, $-NH_2$) react to form a link with loss of a small molecule (water, HCl).
- Polymer molecular weight rises slowly, only at high conversion.
- Eg: nylon-6,6 (hexamethylene diamine + adipic acid), polyester

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(PET), polycarbonate, epoxy, polyurethane, phenol-formaldehyde (bakelite).

| Addition | Condensation |
|---|---|
| Chain reaction; fast | Step reaction; slow |
| No by-product | Small molecules ($H_2O$, HCl) released |
| Monomer has double bond | Monomer has two functional groups |
| High molecular weight at once | MW increases gradually |
| PE, PVC, PS, PMMA | Nylon, PET, bakelite |

## Techniques of polymerisation (industrial)

| Technique | Description | Product |
|---|---|---|
| Bulk (mass) | Monomer + initiator only; heat removal difficult | Pure polymer, PMMA sheets |
| Solution | Monomer in solvent, easy heat control | Adhesives, coatings |
| Suspension | Monomer droplets in water (stabiliser) | Beads: PVC, PS |
| Emulsion | Monomer + surfactant in water; latex | Paints, rubber, PVC paste |

## Copolymerisation and cross-linking

- **Copolymer** : two different monomers (random, alternating, block, graft) → new properties, e.g. P(VDF-TrFE), ABS, SBR.
- **Cross-linking / vulcanisation** : links between chains (sulphur in rubber) → thermosets and elastomers.

## Processing of polymers (shaping)

- **Injection moulding** , extrusion, blow moulding, compression moulding, thermoforming, casting, spin coating (films), 3D printing.

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### Smart polymers - special synthesis

- PVDF is stretched (mechanically drawn) to convert alpha to beta phase (all-trans, polar) and poled (electric field 50-100 MV/m) to become piezoelectric.
- Conducting polymers (polyaniline, polypyrrole, PEDOT): chemical / electrochemical oxidative polymerisation, doped for conductivity.
- Hydrogels : cross-linked hydrophilic polymers (PNIPAM) - swell / shrink with temperature or pH.
- Shape memory polymers : polyurethane segmented block copolymers (hard and soft segments).
- Ionic polymer-metal composite (IPMC) : Nafion membrane + chemically plated Pt electrodes.
- Electroactive polymer (EAP) / dielectric elastomer : acrylic or silicone film between compliant electrodes.

## 3.5 UV Radiation Curing of Polymers

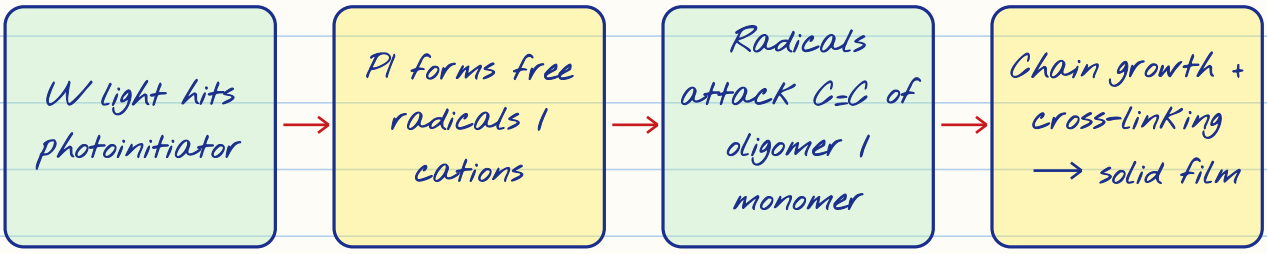
Definition: UV curing is a photochemical process in which high-intensity ultraviolet light instantly converts a liquid monomer / oligomer formulation into a solid cross-linked polymer (photopolymerisation).

### Formulation (UV curable resin)

- Oligomer / prepolymer (epoxy acrylate, urethane acrylate, polyester acrylate) - gives main properties.
- Reactive monomer / diluent (acrylates, eg. TPGDA) - adjusts viscosity and cross-link density.
- Photoinitiator (PI) - absorbs UV (250-400 nm) and generates radicals / acids.
- Additives - stabilisers, pigments, fillers, flow agents.

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## Mechanism




- **Free-radical curing** : acrylates; fast but oxygen inhibition at the surface.
- **Cationic curing** : epoxies, vinyl ethers; no oxygen inhibition, low shrinkage, **dark-cure** continues after light stops; sensitive to moisture.
- **UV sources** : mercury vapour lamps (medium pressure), metal halide lamps, **UV LEDs** (365, 385, 395, 405 nm), excimer lamps.
- Important parameters: wavelength, **intensity** (W/cm2) , **dose** (J/cm2) , exposure time, film thickness, temperature.

## Advantages

- **Very fast** (seconds), room temperature, solvent free (low VOC / eco-friendly), energy efficient, small floor space, good chemical and scratch resistance, **spatial control** of cure (patterns).

## Limitations

- Limited curing depth (shadow areas not cured), oxygen inhibition, shrinkage and stress, cost of photoinitiator and equipment, safety (UV / ozone).

## Applications in smart materials and devices

- **Photoresists** (SU-8 for MEMS), **photo-patterning** of polymer sensors and actuators.
- **Stereolithography** (SLA / DLP) 3D printing of smart structures and 4D printing.
- Coatings on optical fibres (fibre-optic sensors), adhesives for

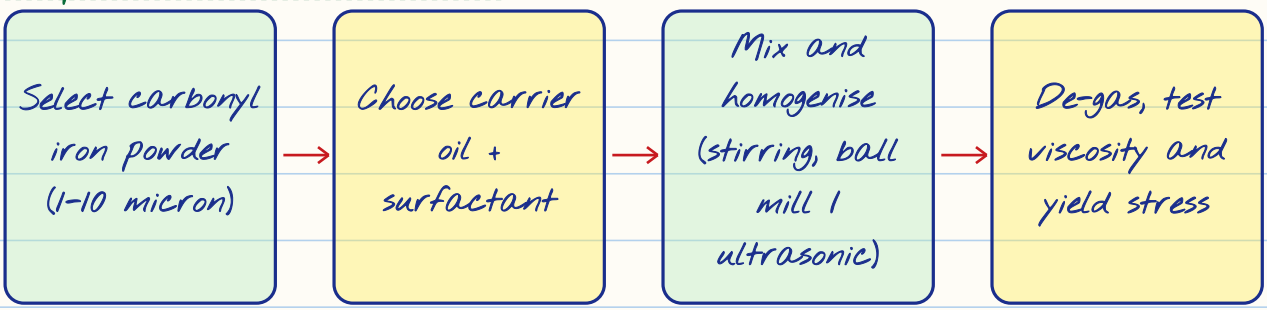
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bonding piezo patches, encapsulation of sensors, dental fillings, printing inks, protective coatings, curing of polymer films and IPMC / dielectric elastomer electrodes.

## 3.6 Processing of Smart Fluids (ER, MR fluids and ferrofluid

**Definition:** Smart fluids are colloidal suspensions of fine particles in a carrier liquid; processing means selecting and preparing particles, carrier fluid and additives and then mixing and stabilising them.

### Preparation of MR fluid




- Particles are carbonyl iron (from $Fe(CO)_5$ decomposition), iron-cobalt, volume fraction 20–40%.
- Carrier fluids: silicone oil, mineral oil, synthetic hydrocarbon, water, glycol – low viscosity and wide temperature range.
- Additives: surfactants (oleic acid), thixotropic agents (fumed silica, organoclay), anti-wear and anti-oxidant – prevent sedimentation, agglomeration.
- Coating of particles (silica, polymer) improves stability.

### Preparation of ER fluid

- Dielectric particles (silica, titanium dioxide, starch, zeolite, polyaniline, barium titanate) are dried (to control moisture), dispersed in silicone oil with stabiliser, and ball-milled.
- Giant ER fluids – nano-coated $TiO_2$ particles with urea give

yield stress > 100 kPa.

- Types: dry / water-free ER, polymer-based and composite particles.

### Ferrofluids

- Stable colloid of magnetic nano-particles ($Fe_3O_4$, ~10 nm) coated with surfactant in oil / water; made by co-precipitation of $Fe^{2+}$ and $Fe^{3+}$ salts with ammonia.
- Does not form chains but is attracted to a magnet (liquid seals, speakers, heat transfer).

### Quality aspects

- Particle size and shape, density mismatch (iron 7.8 vs oil 0.95 g/cc → settling), shear stability, viscosity at zero field, temperature range, off-state viscosity, durability.

> Note: Sedimentation is reduced by using nano-sized particles, density-matching hollow particles, thixotropic additives and magnetic-gel / MR elastomer forms.

## 3.7 Quick Revision / Important Questions

- 1. Describe the processing of silicon from sand to a wafer (CZ method).
- 2. Explain photolithography and the steps of microfabrication.
- 3. What is metallization? Explain evaporation, sputtering and electroplating.
- 4. Explain the solid-state processing of PZT ceramics with a flow chart.
- 5. Differentiate addition and condensation polymerisation with examples.
- 6. What is UV curing? Explain its mechanism, advantages and applications.

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- 7. How are MR / ER fluids prepared? Write the role of additives.

IMPORTANT: Exam tip: draw the flow chart for each process (Si wafer, ceramic route, UV curing) - flow charts score marks quickly.